

One-Dimensional Optimization

Learning objectives

After studying this chapter, you should be able to

- describe the common structure, stopping tests, costs, and convergence questions of an iterative optimization algorithm;
- explain how line search reduces a multivariate problem to a one-dimensional problem;
- implement bisection on the derivative and prove its worst-case bracket complexity;
- derive Newton's iteration from a quadratic Taylor model and prove local quadratic convergence;
- derive golden-section search, including the deletion rule and the golden ratio; and
- choose among the three methods according to the available information and the required reliability.

Chapter 6 developed necessary conditions that identify stationary candidates, and Chapter 7 developed second-order conditions that can certify their local behavior. Except for special functions, however, the stationarity equation cannot be solved in closed form. Optimization algorithms therefore construct a sequence of numerical approximations. This chapter first records the common algorithmic framework in arbitrary dimension and then studies three one-dimensional methods. These methods isolate choices that recur later: derivatives versus function values, fast local steps versus reliable brackets, and exact versus inexact solution of a line-search subproblem.

8.1 From optimality conditions to an iterative algorithm

Let $n \in \mathbb{N}$, let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, and let $f : \mathcal{D} \rightarrow \mathbb{R}$ be differentiable. A generic iterative method starts from $\mathbf{x}^{(0)} \in \mathcal{D}$ and, at iteration k , uses the available history

$$\mathcal{H}^{(k)} := \{\mathbf{x}^{(0)}, \dots, \mathbf{x}^{(k)}; f, \nabla f, \mathbf{H}_f \text{ values already computed}\}$$

to generate a new point $\mathbf{x}^{(k+1)}$. It then evaluates one or more stopping tests. Thus the common loop is

initialize $\longrightarrow$ generate a trial iterate $\longrightarrow$ test and either stop or continue.

This description is independent of the dimension and of the particular rule used to generate the next iterate.

8.1.1 Line search and trust regions

A line-search method separates the decision into a direction and a length:

$$\mathbf{x}^{(k+1)} := \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \alpha^{(k)} \geq 0. \quad (8.1)$$

The direction $\mathbf{d}^{(k)}$ may be a steepest-descent, conjugate-gradient, Newton, or another model-based direction. Once the direction is fixed, define the restriction

$$\phi^{(k)}(\alpha) := f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}), \quad \mathcal{I}^{(k)} := \{\alpha \geq 0 : \mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)} \in \mathcal{D}\}. \quad (8.2)$$

Choosing the step length is therefore a one-dimensional optimization problem. An *exact* line search finds a member of $\arg \min_{\alpha \in \mathcal{I}^{(k)}} \phi^{(k)}(\alpha)$, when this set is nonempty. An *inexact* line search merely asks for a step satisfying a prescribed decrease test (more details in [Chapter 9](#)).

A trust-region method instead chooses direction and length simultaneously by approximately minimizing a model over a neighborhood such as $\{\mathbf{d} : \|\mathbf{d}\| \leq \Delta^{(k)}\}$. We mention this alternative to locate line search in the broader algorithmic picture; trust-region algorithms are not developed in this course.

Warning

The one-dimensional restriction does not remove the assumptions of the method used to solve it. An exact line-search minimizer may fail to exist; a zero of $(\phi^{(k)})'$ need not minimize $\phi^{(k)}$; and a bracketing method requires a valid bracket.

8.1.2 Stopping tests

No finite computation can normally verify an exact equality such as $\nabla f(\mathbf{x}^{(k)}) = \mathbf{0}$. Let $\varepsilon_g, \varepsilon_x, \varepsilon_f > 0$. For each residual one may use an absolute test or its scaled counterpart:

$$\|\nabla f(\mathbf{x}^{(k)})\| \leq \varepsilon_g \quad \text{or} \quad \|\nabla f(\mathbf{x}^{(k)})\| \leq \varepsilon_g \max\{1, \|\nabla f(\mathbf{x}^{(0)})\|\}, \quad (8.3)$$

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\| \leq \varepsilon_x \quad \text{or} \quad \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\| \leq \varepsilon_x \max\{1, \|\mathbf{x}^{(k)}\|\}, \quad (8.4)$$

$$|f(\mathbf{x}^{(k+1)}) - f(\mathbf{x}^{(k)})| \leq \varepsilon_f \quad \text{or} \quad |f(\mathbf{x}^{(k+1)}) - f(\mathbf{x}^{(k)})| \leq \varepsilon_f \max\{1, |f(\mathbf{x}^{(k)})|\}. \quad (8.5)$$

If the optimal value f^* or an optimizer $\mathbf{x}^*$ is known for a test problem, one may instead monitor

$$\begin{aligned} |f(\mathbf{x}^{(k)}) - f^*| &\leq \varepsilon_f \quad \text{or} \quad |f(\mathbf{x}^{(k)}) - f^*| \leq \varepsilon_f \max\{1, |f^*|\}, \\ \|\mathbf{x}^{(k)} - \mathbf{x}^*\| &\leq \varepsilon_x \quad \text{or} \quad \|\mathbf{x}^{(k)} - \mathbf{x}^*\| \leq \varepsilon_x \max\{1, \|\mathbf{x}^*\|\}. \end{aligned}$$

A maximum iteration count $K_{\max}$ and a maximum running time $T_{\max}$ are safeguards rather than evidence of success.

Warning

A small gradient certifies only approximate stationarity, not local minimality. A small step or objective change can also indicate stagnation caused by poor scaling, an inaccurate direction, or an excessively small step length. A reliable implementation records which test caused termination and preferably uses a stationarity test together

with a stagnation safeguard.

8.1.3 Cost and convergence questions

The cost of an iteration includes the number of function, gradient, and Hessian evaluations, the linear-algebra work, and the memory used. Given a particular stopping condition, *total complexity* counts the combined work needed to meet it. Counting only outer iterations can therefore be misleading: one iteration involving a Hessian factorization or many trial function values may cost much more than a simple gradient step. Hardware, parallelism, sparsity, and memory traffic can change the practical comparison further.

Convergence statements must specify both the problem class and the allowed initializations. A local theorem assumes that $\mathbf{z}^{(0)}$ is sufficiently near a target; a global convergence theorem covers every allowed initialization in the stated problem class. In numerical optimization, the phrase *globally convergent* often means convergence to stationary points or that every limit point is stationary—not convergence to a global minimizer.

Definition 8.1 (Q-rates). Let $\mathbf{z}^{(k)} \rightarrow \mathbf{z}^*$ and put $e^{(k)} := \|\mathbf{z}^{(k)} - \mathbf{z}^*\|$. The convergence is

- *Q-linear* if there are $q \in (0, 1)$ and K such that $e^{(k+1)} \leq qe^{(k)}$ for every $k \geq K$;
- *Q-superlinear* if $e^{(k+1)}/e^{(k)} \rightarrow 0$, with ratios taken when $e^{(k)} > 0$; and
- *Q-quadratic* if there are $C > 0$ and K such that $e^{(k+1)} \leq C(e^{(k)})^2$ for every $k \geq K$.

There is no single method that is uniformly best for all objectives, dimensions, derivative availability, starting points, accuracy requirements, and computing platforms. The assumptions and cost model are part of every meaningful algorithm comparison.

8.2 A common one-dimensional test problem

The three methods below will be illustrated on

$$f(x) := e^x + x^2 - 2x, \quad x \in \mathcal{D} = \mathbb{R}. \quad (8.6)$$

Since

$$f'(x) = e^x + 2x - 2, \quad f''(x) = e^x + 2 > 0,$$

the function is strictly convex. Its unique stationary point is therefore its unique global minimizer on $\mathbb{R}$. Numerically,

$$x^* \approx 0.314923058, \quad f^* \approx 0.839484301. \quad (8.7)$$

For the two bracketing methods we use the initial interval $[-1, 1]$. Using one objective makes the methods' information requirements and progress directly comparable.

8.3 Bisection on the derivative

Suppose $f : \mathcal{D} \rightarrow \mathbb{R}$, where $\mathcal{D} \subseteq \mathbb{R}$ is open, and $[a^{(0)}, b^{(0)}] \subset \mathcal{D}$. Bisection applies the intermediate value theorem to f' .

Theorem 8.1 (A derivative bracket). *Suppose f' is continuous on $[a^{(0)}, b^{(0)}]$ and*

$$f'(a^{(0)})f'(b^{(0)}) < 0.$$

Then there is an $x^ \in (a^{(0)}, b^{(0)})$ such that $f'(x^*) = 0$.*

Proof. The strict sign change places zero strictly between the two endpoint values of the continuous function f' . The intermediate value theorem gives a point $x^* \in (a^{(0)}, b^{(0)})$ with $f'(x^*) = 0$. $\square$

Algorithm: Bisection for a stationary point

Choose $a^{(0)} < b^{(0)}$ with $f'(a^{(0)})f'(b^{(0)}) \leq 0$, tolerances $\varepsilon_x > 0$, $\varepsilon_g \geq 0$, and a maximum iteration count. Return an endpoint immediately if its derivative satisfies the stationarity tolerance, and return the initial midpoint without another derivative evaluation if its half-width is already at most ε_x . For $k = 0, 1, \dots, K_{\max} - 1$:

1. Set $x^{(k)} := (a^{(k)} + b^{(k)})/2$ and evaluate $f'(x^{(k)})$.
2. If $|f'(x^{(k)})| \leq \varepsilon_g$, return $x^{(k)}$ as approximately stationary.
3. Retain the half bracket whose endpoint derivative values have opposite signs, calling it $[a^{(k+1)}, b^{(k+1)}]$.
4. If $b^{(k+1)} - a^{(k+1)} \leq 2\varepsilon_x$, return its midpoint.

If all $K_{\max}$ reductions are used without meeting a tolerance, report that the iteration limit was reached; do not label the last midpoint as a successful return.

For Equation (8.6), $f'(-1) < 0 < f'(1)$. The first midpoint and derivative values are

k	0	1	2	3
$x^{(k)}$	0	0.5	0.25	0.375
$f'(x^{(k)})$	-1	0.648721	-0.215975	0.204991

Thus the successive retained brackets begin

$$[-1, 1] \supset [0, 1] \supset [0, 0.5] \supset [0.25, 0.5] \supset [0.25, 0.375] \supset \dots$$

Strict convexity, not bisection alone, identifies their target as the unique global minimizer in Equation (8.7).

Theorem 8.2 (Bisection contraction and complexity). *Assume the hypotheses of Theorem 8.1, and suppose each update retains a sign-changing half interval. After k reductions,*

$$\ell^{(k)} := b^{(k)} - a^{(k)} = 2^{-k} \ell^{(0)}. \quad (8.8)$$

The nested brackets have a unique common point r , this point satisfies $f'(r) = 0$, and the midpoint $x^{(k)}$ of the k -th bracket satisfies

$$|x^{(k)} - r| \leq \frac{\ell^{(k)}}{2}.$$

Consequently,

$$N_{\text{bis}} := \max \left\{ 0, \left\lceil \log_2 \left(\frac{\ell^{(0)}}{2\varepsilon_x} \right) \right\rceil \right\}$$

reductions suffice for the location tolerance ε_x . The bracket widths converge Q-linearly with factor $1/2$.

Proof. One update replaces the current interval by one of its two halves, so $\ell^{(k+1)} = \ell^{(k)}/2$. Induction proves Equation (8.8). The retained endpoint derivatives have opposite signs, so Theorem 8.1 applies at every iteration. Moreover, the closed intervals are nested and their lengths tend to zero. The nested-interval theorem therefore gives

$$\bigcap_{k=0}^{\infty} [a^{(k)}, b^{(k)}] = \{r\}$$

for a unique point r . For every k , choose a root $r^{(k)}$ of f' in the k -th bracket. Then $|r^{(k)} - r| \leq \ell^{(k)} \rightarrow 0$, and continuity of f' yields $f'(r) = 0$. The midpoint bound follows because both $x^{(k)}$ and r belong to the same interval. Solving $2^{-k}\ell^{(0)} \leq 2\varepsilon_x$ for the least nonnegative integer k gives the stated bound. Finally, $\ell^{(k+1)}/\ell^{(k)} = 1/2$, which is Q-linear convergence of the widths. $\square$

Warning

The theorem guarantees a root of f' , not a minimizer. To infer a strict local minimum from signs, one needs $f' < 0$ throughout a left neighborhood of the root and $f' > 0$ throughout a right neighborhood; two endpoint signs alone do not suffice. For example, for $f(x) = -x^2$ on $[-1, 1]$, the endpoint derivatives have opposite signs and bisection immediately returns 0, which is the global maximum. The Q-linear statement above concerns bracket widths; the midpoint errors have the displayed geometric bound but need not themselves have a Q-linear error ratio.

After the two endpoint evaluations, each reduction needs one new derivative evaluation, a sign comparison, and constant scalar storage.

8.4 One-dimensional Newton method

At $x^{(k)}$, form the quadratic Taylor model

$$q^{(k)}(x) := f(x^{(k)}) + f'(x^{(k)})(x - x^{(k)}) + \frac{1}{2}f''(x^{(k)})(x - x^{(k)})^2. \quad (8.9)$$

Its stationarity equation is $f'(x^{(k)}) + f''(x^{(k)})(x - x^{(k)}) = 0$. When $f''(x^{(k)}) \neq 0$, this gives

$$d^{(k)} = -\frac{f'(x^{(k)})}{f''(x^{(k)})}, \quad x^{(k+1)} = x^{(k)} + d^{(k)}. \quad (8.10)$$

This is Newton's root-finding method applied to $f'(x) = 0$. If $f''(x^{(k)}) > 0$, $x^{(k+1)}$ is the unique global minimizer of $q^{(k)}$; if $f''(x^{(k)}) < 0$, it is the unique global maximizer. If $f''(x^{(k)}) = 0$, the Newton quotient is undefined. When also $f'(x^{(k)}) = 0$, a stationarity test should stop the algorithm before that quotient is formed.

Algorithm: One-dimensional Newton method

Choose $x^{(0)} \in \mathcal{D}$, a stationarity tolerance $\varepsilon_g > 0$, a step tolerance $\varepsilon_x > 0$, and a maximum of $K_{\max}$ Newton steps. For $k = 0, 1, \dots, K_{\max}$:

1. Evaluate $f'(x^{(k)})$. If $|f'(x^{(k)})| \leq \varepsilon_g$, return $x^{(k)}$.
2. If $k = K_{\max}$, report that the iteration limit was reached.
3. Evaluate $f''(x^{(k)})$. If it is zero or numerically too small, stop or switch to a safeguarded method.
4. Form $d^{(k)} = -f'(x^{(k)})/f''(x^{(k)})$. If $|d^{(k)}| \leq \varepsilon_x \max\{1, |x^{(k)}|\}$ while the stationarity test has failed, report stagnation.
5. Set $x^{(k+1)} = x^{(k)} + d^{(k)}$; stop with a diagnostic if the trial point leaves $\mathcal{D}$.

For Equation (8.6), Newton's method from $x^{(0)} = 0$ with $\varepsilon_g = 10^{-8}$ gives

k	$x^{(k)}$	$f(x^{(k)})$	$f'(x^{(k)})$
0	0.000000000	1.000000000	-1.000×10^0
1	0.333333333	0.840056870	6.228×10^{-2}
2	0.314992285	0.839484309	2.333×10^{-4}
3	0.314923059	0.839484301	3.283×10^{-9}

The last estimate has location error approximately 9.74×10^{-10} .

Theorem 8.3 (Local quadratic convergence in one dimension). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be twice differentiable on an open interval $\mathcal{D}$, and let $x^* \in \mathcal{D}$ satisfy*

$$f'(x^*) = 0, \quad f''(x^*) \neq 0.$$

Suppose f'' is Lipschitz continuous near x^ . Then there are $\delta > 0$ and $C > 0$ such that, whenever $|x^{(0)} - x^*| < \delta$, the Newton iterates are well defined, remain near x^* , and satisfy*

$$|x^{(k+1)} - x^*| \leq C |x^{(k)} - x^*|^2.$$

Thus the convergence is Q-quadratic.

Proof. Put $g := f'$. Continuity of $g' = f''$ and $g'(x^*) \neq 0$ give a closed interval $U \subset \mathcal{D}$, centered at x^* , and a number $m > 0$ such that $|g'(x)| \geq m$ for every $x \in U$. Let L be a Lipschitz constant for g' on U . For a current iterate $x^{(k)} \in U$, write $e^{(k)} := x^{(k)} - x^*$. The fundamental theorem of calculus along the segment from x^* to $x^{(k)}$ gives

$$g(x^{(k)}) - g(x^*) = e^{(k)} \int_0^1 g'(x^* + te^{(k)}) dt.$$

Since $g(x^*) = 0$, the Newton update gives, one equality at a time,

$$\begin{aligned} e^{(k+1)} &= x^{(k)} - x^* - \frac{g(x^{(k)})}{g'(x^{(k)})} \\ &= e^{(k)} - \frac{g(x^{(k)}) - g(x^*)}{g'(x^{(k)})} \end{aligned}$$

$$\begin{aligned}
 &= \frac{g'(x^{(k)})e^{(k)} - g(x^{(k)})}{g'(x^{(k)})} \\
 &= \frac{e^{(k)}}{g'(x^{(k)})} \int_0^1 [g'(x^{(k)}) - g'(x^* + te^{(k)})] dt.
 \end{aligned}$$

Therefore

$$|e^{(k+1)}| \leq \frac{L}{m} |e^{(k)}|^2 \int_0^1 (1-t) dt = \frac{L}{2m} |e^{(k)}|^2.$$

Choose any $C > 0$ with $C \geq L/(2m)$, and choose $\delta > 0$ so that $[x^* - \delta, x^* + \delta] \subset U$ and $C\delta < 1$. If $e^{(k)} = 0$, the stationarity test stops, or the exact Newton map leaves the iterate at x^* . Otherwise, whenever $|e^{(k)}| < \delta$, the preceding inequality gives $|e^{(k+1)}| < |e^{(k)}| < \delta$. Induction proves that every iterate remains in U , so the update is well defined and the quadratic bound holds for all k . $\square$

Each iteration uses one first- and one second-derivative evaluation and scalar arithmetic, with constant scalar storage in one dimension. The local theorem explains the rapid final convergence, but it does not give a worst-case iteration bound from an arbitrary starting point.

Example 8.1 (Nearby initial points, different Newton behavior). Let

$$h(x) := \sin x - 0.8x + 0.05x^2.$$

Newton's method for $h'(x) = 0$ is

$$x^{(k+1)} = x^{(k)} - \frac{\cos x^{(k)} - 0.8 + 0.1x^{(k)}}{-\sin x^{(k)} + 0.1}.$$

From $x^{(0)} = 0.08$, the first iterates are

$$0.08, -10.1165938, -14.9159483, -11.2277027, -13.1662754, \dots,$$

and the sequence approaches the two-cycle

$$-13.1189770 \longleftrightarrow -11.1014970.$$

In contrast, from $x^{(0)} = 0.12$, the iterates begin

$$0.12, 10.5099392, 10.7291396, 10.7201040, \dots$$

and converge to the local minimizer $x^* \approx 10.7200938$. The two starts straddle

$$0.08 < \arcsin(0.1) \approx 0.1001674 < 0.12, \quad h''(\arcsin(0.1)) = 0.$$

At the reported limit, $|h'(x^*)| \approx 4.1 \times 10^{-8}$ and $h''(x^*) \approx 1.0623 > 0$. Near the intervening zero of h'' , the Newton denominator is arbitrarily small, so a small change in the initial point can produce a very different first step.

Warning

For Newton's method to run from a given initial point, the second derivative must be nonzero at every nonterminal iterate and every trial point must remain in $\mathcal{D}$. The

weaker condition $f''(x^*) \neq 0$, together with local regularity, is enough only for a local theorem with a sufficiently close initialization. Newton may converge to a local maximum or a nonextremal stationary point, cycle, or leave the domain.

8.5 Golden-section search

Golden-section search uses only function values. Its shape assumption replaces derivative information.

Definition 8.2 (Strict unimodality for minimization). Let $a < b$. A function $f : [a, b] \rightarrow \mathbb{R}$ is *strictly unimodal* if there is a unique $x^* \in [a, b]$ such that f is strictly decreasing on $[a, x^*]$ and strictly increasing on $[x^*, b]$, with either condition vacuous when x^* is an endpoint.

Differentiability is not required. For example, $f(x) = |x|$ is strictly unimodal on $[-1, 1]$: it is strictly decreasing on $[-1, 0]$, strictly increasing on $[0, 1]$, and has the unique minimizer $x^* = 0$.

Lemma 8.4 (Golden-section deletion rule). *Let f be strictly unimodal on $[a, b]$ with minimizer x^* , and let $a < x_1 < x_2 < b$. Then*

$$\begin{aligned} f(x_1) > f(x_2) &\implies x^* \in (x_1, b], \\ f(x_1) < f(x_2) &\implies x^* \in [a, x_2), \\ f(x_1) = f(x_2) &\implies x^* \in (x_1, x_2). \end{aligned}$$

Proof. If $f(x_1) > f(x_2)$ and $x^* \leq x_1$, both points lie to the right of the minimizer, where strict increase would imply $f(x_1) < f(x_2)$, a contradiction. Hence $x^* > x_1$. The second implication follows symmetrically. If the values are equal and $x^* \leq x_1$, strict increase gives $f(x_1) < f(x_2)$; if $x^* \geq x_2$, strict decrease gives $f(x_1) > f(x_2)$. Both alternatives are impossible, so $x^* \in (x_1, x_2)$. $\square$

Proposition 8.5 (The reusable ratio). *Suppose that, in every interval $[a, b]$ of length ℓ , two interior points are placed symmetrically at*

$$a + (1 - \rho)\ell \quad \text{and} \quad a + \rho\ell, \quad \rho \in (1/2, 1).$$

Require every retained side interval to have length $\rho\ell$, with one old interior point occupying the corresponding interior position after rescaling. Then

$$\rho^2 = 1 - \rho, \quad \rho = \frac{\sqrt{5} - 1}{2}.$$

Proof. For an interval of length ℓ , place the interior points at distances $(1 - \rho)\ell$ and $\rho\ell$ from the left endpoint. Suppose the left part is discarded. The retained interval has length $\rho\ell$, and the old right interior point is at distance $(2\rho - 1)\ell$ from its new left endpoint. To reuse it as the new left interior point, this distance must equal $(1 - \rho)(\rho\ell)$. Cancelling ℓ yields $2\rho - 1 = \rho - \rho^2$, or $\rho^2 = 1 - \rho$. The roots are $(-1 \pm \sqrt{5})/2$, and the unique root in $(1/2, 1)$ is the stated value. The case in which the right part is discarded is symmetric. $\square$

The conventional golden ratio is $\varphi := (1 + \sqrt{5})/2$, and the contraction factor used by the algorithm is its reciprocal $\rho := \varphi^{-1} = (\sqrt{5} - 1)/2$. In a current bracket $[a^{(k)}, b^{(k)}]$, define

$$\lambda^{(k)} := b^{(k)} - \rho(b^{(k)} - a^{(k)}), \quad \mu^{(k)} := a^{(k)} + \rho(b^{(k)} - a^{(k)}).$$

Algorithm: Golden-section search

Choose a strict-unimodality bracket $[a^{(0)}, b^{(0)}]$, a location tolerance $\varepsilon_x > 0$, and a maximum of $K_{\max}$ reductions. If the initial width is at most $2\varepsilon_x$, return its midpoint. Otherwise form the two golden interior points and evaluate f there. For $k = 0, 1, \dots, K_{\max} - 1$:

1. Use [Lemma 8.4](#) to retain an interval containing the minimizer. If the two values are equal, retain either one of the two length- ρ side intervals; both contain the minimizer.
2. Reuse the surviving interior point, form the other golden interior point, and make exactly one new function evaluation.
3. If the new width is at most $2\varepsilon_x$, return its midpoint.

If all $K_{\max}$ reductions are used without meeting the location tolerance, report that the iteration limit was reached.

For [Equation \(8.6\)](#) on $[-1, 1]$, the initial points are

$$\lambda^{(0)} = -0.236068, \quad \mu^{(0)} = 0.236068,$$

and

$$f(\lambda^{(0)}) = 1.317591 > 0.849853 = f(\mu^{(0)}).$$

The first retained bracket is therefore $[-0.236068, 1]$. Reusing 0.236068, the new point is 0.527864, and

$$f(0.236068) = 0.849853 < 0.918220 = f(0.527864).$$

Hence the second retained bracket is $[-0.236068, 0.527864]$.

Theorem 8.6 (Golden-section contraction and complexity). *Let f be strictly unimodal on the initial bracket, and suppose golden-section search performs k reductions. Then the retained bracket contains the unique minimizer and has length*

$$\ell^{(k)} = \rho^k \ell^{(0)}. \tag{8.11}$$

Consequently,

$$N_{\text{gold}} := \max \left\{ 0, \left\lceil \frac{\log(\ell^{(0)}/(2\varepsilon_x))}{\log(1/\rho)} \right\rceil \right\}$$

reductions suffice for midpoint location error at most ε_x . The bracket widths converge Q -linearly with factor ρ , and, after the two initial values, each reduction requires one new function value. If $N_{\text{gold}} = 0$, the initial midpoint already satisfies the location tolerance and no function value is needed; otherwise the total is $N_{\text{gold}} + 2$ function values for the displayed implementation.

Proof. The deletion lemma shows inductively that every retained interval contains x^* . By construction, one update retains a fraction ρ of the current length; induction yields [Equation \(8.11\)](#). The midpoint of an interval containing x^* is at distance at most half its width

from x^* . Solving $\rho^k \ell^{(0)} \leq 2\varepsilon_x$ gives the stated least-integer bound because $\log(1/\rho) > 0$. The identity $\ell^{(k+1)}/\ell^{(k)} = \rho$ proves Q-linear convergence of the widths. Finally, [Proposition 8.5](#) guarantees that one interior point and its value are reused, so only one new value is required per reduction after initialization. The algorithm's initial-width test gives the stated zero-call case; when at least one reduction is required, two initial values followed by one new value per reduction give $N_{\text{gold}} + 2$ calls. $\square$

The last new value is not needed if the location test succeeds immediately after the final deletion. An implementation that tests the new width before forming the next interior point therefore uses $N_{\text{gold}} + 1$ values when $N_{\text{gold}} \geq 1$. The Python implementation used for the comparison below follows the displayed pseudocode and reports $N_{\text{gold}} + 2$.

Applied to [Equation \(8.6\)](#) on $[-1, 1]$, golden-section search uses no derivatives and produces nested brackets converging to x^* . It contracts the bracket more slowly than derivative bisection because $\rho \approx 0.618 > 1/2$, but it needs only objective values.

Warning

The shape assumption is essential; even a unique global minimizer is not enough. On $[0, 1]$, let

$$f(x) = \min\{(x - 0.1)^2, 0.01 + (x - 0.8)^2\}.$$

Its unique global minimizer is 0.1, but for the first golden points $\lambda \approx 0.382$ and $\mu \approx 0.618$, one has $f(\lambda) > f(\mu)$. The deletion rule would discard $[0, \lambda]$, including the global minimizer, because this function is not strictly unimodal. As with bisection, the unconditional Q-linear statement concerns bracket width, not necessarily the returned midpoint sequence.

8.6 Comparison and method selection

Method	Main assumptions	Per reduction/iteration	Guaranteed rate
Derivative bisection	Continuous f' , sign-changing bracket; extra condition needed for minimality	One new f' value, constant storage	Bracket width Q-linear, factor $1/2$; explicit worst-case bound
Newton	f', f'' ; nonzero f'' along the run; local regularity near target	One f' , one f'' , scalar solve; constant scalar storage	Locally Q-quadratic near a regular root; no unsafeguarded global bound
Golden section	Strict unimodality and a containing bracket	Two initial f values, then one new value; constant storage	Bracket width Q-linear, factor $\rho \approx 0.618$; explicit worst-case bound

For the common strictly convex function, all three methods approach the same point. Their raw iteration counts are not directly comparable because they use different oracles: bisection calls f' , Newton calls both f' and f'' , and golden section calls only f . Newton is typically

fastest once sufficiently close; the bracketing methods trade speed for a retained location guarantee.

For a reproducible comparison, use $[-1, 1]$, location tolerance $\varepsilon_x = 10^{-4}$ for both bracketing methods, and $\varepsilon_g = 10^{-8}$ for Newton starting from $x^{(0)} = 0$. The corresponding Python implementations give

method	updates	estimate	location error	oracle calls
bisection on f'	14	0.314880371	4.269×10^{-5}	16 f'
Newton	3	0.314923059	9.742×10^{-10}	4 f' , 3 f''
golden section	20	0.314935576	1.252×10^{-5}	22 f

Here an update means one bracket reduction for a bracketing method and one Newton step for Newton's method. Bisection counts its two endpoint derivative values; golden section counts its two initial interior values. The reference value used for the displayed errors is $x^* = 0.314923057845$. These conventions explain the exact counts and must be stated whenever implementations are compared.

Chapter summary

- An optimization method must specify initialization, iterate generation, stopping diagnostics, computational cost, and a convergence statement.
- Line search separates direction from step length and creates a one-dimensional subproblem; trust regions choose both together.
- Derivative bisection is reliable within a sign-changing bracket but finds a stationary point, not automatically a minimizer.
- Newton is locally Q-quadratic at a regular stationary point, but the unsafeguarded method has no comparable guarantee from a remote start.
- Golden-section search is derivative-free and reuses one function value, but relies on unimodality.
- No method dominates across all problem classes, information costs, and computing platforms.

8.7 Exercises

The exercises are ordered by topic and, within each topic, broadly by difficulty.

Exercise 8.1 (Reading stopping criteria). Suppose $f(x) := (x - 2)^2$, $x^{(0)} = 2 + 10^{-4}$, and $x^{(1)} = 2 + 10^{-6}$. With $\varepsilon_g = \varepsilon_x = \varepsilon_f = 10^{-4}$, test both the absolute and scaled versions of Equations (8.3) to (8.5). State what each satisfied test does and does not certify.

Exercise 8.2 (Two bisection reductions). For $f(x) := x^4 + 2x$, apply bisection to f' on $[-1, 1]$ for two interval reductions. Report the midpoints, derivative signs, and final bracket.

Exercise 8.3 (Two Newton iterations). For $f(x) := x^4 + 2x$, start Newton's method at $x^{(0)} := -1$. Compute $x^{(1)}$ and $x^{(2)}$ exactly and verify every derivative used.

Exercise 8.4 (Taylor model and a full Newton step). For the family

$$f_c(x) := 2x + x^2 + cx^6, \quad c \in \mathbb{R},$$

form the second-order Taylor model at $x^{(0)} = 0$ and compute the full Newton iterate $x^{(1)}$. Determine exactly for which c this step decreases, leaves unchanged, or increases the objective. Does $f_c''(x^{(0)}) > 0$ alone guarantee a decrease in f_c ?

Exercise 8.5 (Checking unimodality). Let

$$f(x) := \begin{cases} x^2 - 1, & x \geq -1, \\ -x - 1, & x < -1. \end{cases}$$

Determine whether f is strictly unimodal on $[-5, 5]$, and locate its minimizer.

Exercise 8.6 (Two golden-section reductions). For $f(x) := x^2 + 2x$ on $[-1, 1]$, perform two golden-section reductions. Record both interior points and function comparisons at each step, and verify that the final interval contains the boundary minimizer.

Exercise 8.7 (When derivative bisection is invalid). An implementation attempts to use the same sign-bracketing logic to find a zero of

$$h(x) := \frac{1}{x^2 - 4}$$

from $[1/2, 2]$. Check every assumption of [Theorem 8.1](#) and justify your answer.

Exercise 8.8 (A stationary point need not minimize). Let $f(x) := (x^2 - 1)^2$. Apply derivative bisection on $[-1/2, 1/2]$. Determine the point returned immediately, classify it, and explain why a valid sign-changing derivative bracket does not certify a minimizer.

Exercise 8.9 (Comparing worst-case bracket costs). Both bracketing methods start from an interval of length 4. Determine how many reductions each method needs to guarantee midpoint location error at most 10^{-6} . Count the derivative or function evaluations, including the initial endpoint/interior evaluations. If one derivative evaluation costs $c > 0$ times as much as one function evaluation and all other costs are ignored, determine for which c golden-section search has the smaller worst-case oracle cost. Explain why this simplified comparison still need not predict wall-clock time.

Exercise 8.10 (Newton can target the wrong stationary point). Construct a twice continuously differentiable function, different from the functions in the preceding exercises, with one local maximum and two local minima such that Newton's method for $f' = 0$, from a stated *nonstationary* initial point, makes at least three nonconstant updates and converges to the local maximum. Verify the first three updates and explain why $f''(x^*) \neq 0$ controls the local rate while $f''(x^*) > 0$, not merely nonzero curvature, is needed to classify x^* as a local minimum.